

9(a)(i)	(induced) e.m.f. (directly) proportional to rate	M1
	of change of magnetic flux (linkage)	A1
9(a)(ii)	e.m.f. = 0 apart from thin pulses at t_1 and t_2	B1
	rectangular pulses centred on t_1 and t_2 , of widths 2 small squares and 1 small square respectively	B1
	e.m.fs. at t_1 and t_2 have opposite polarities	B1
	magnitude of e.m.f. at t_2 double the magnitude of e.m.f. at t_1	B1
9(b)	V_H shown as zero before ($t_1 - 2$ squares) and after ($t_2 + 2$ squares) and rises to a constant non-zero value between t_1 and t_2	M1
	change at t_1 shown as 2 small squares wide and change at t_2 shown as 1 small square wide	A1